

Canaan Avalon Bitcoin Miner Diagnostic Checklist

Technician Name: _____ Date: _____
Miner Model: _____ Serial Number: _____
Firmware Version: _____ Reported Issue: _____
Customer/Location: _____ Ticket #: _____

1. External Inspection

- ☐ Check for physical damage to case/chassis
- ☐ Inspect power cables for damage or fraying
- ☐ Check APW power supply unit for physical damage
- ☐ Inspect network cable and connection
- ☐ Look for signs of overheating (discoloration, melting)
- ☐ Check for unusual odors (burnt electronics)
- ☐ Inspect fans for dust accumulation or damage
- ☐ Verify all screws and fasteners are present
- ☐ Inspect air intake and exhaust areas for blockage
- ☐ Check for any liquid damage or residue
- ☐ Note any modification or non-standard components

2. Power Supply Assessment

- ☐ Verify APW power supply model and compatibility
- ☐ Check power input connections
- ☐ Measure power supply output voltages with multimeter:
 - ☐ Main voltage: _____ V (Target: typically 11.8-12.5V)
 - ☐ Other voltage rails (if applicable): _____ V
- ☐ Check PSU fan functionality
- ☐ Inspect PSU internal components if suspect (qualified personnel only)
- ☐ Record PSU model/rating: _____
- ☐ Take photos of any damage or abnormalities
- ☐ Check PCI-E power connectors for burns or damage

3. Internal Hardware Inspection

- ☐ Remove covers following proper ESD procedures
- ☐ Inspect control board for:
 - ☐ Burnt or damaged components
 - ☐ Bulging or leaking capacitors
 - ☐ Corrosion or residue
 - ☐ Cold solder joints
 - ☐ Broken traces
 - ☐ AUC controller condition (if applicable)
 - ☐ Ethernet port condition
- ☐ Inspect all hash boards for:
 - ☐ Burnt ASIC chips
 - ☐ Damaged components
 - ☐ Corrosion on connectors
 - ☐ Physical damage to board
 - ☐ Burned traces or areas
 - ☐ Proper installation in slots
 - ☐ Chip array condition (visual)
- ☐ Check all internal connections and cables:
 - ☐ Data cables properly seated
 - ☐ Power connectors secure
 - ☐ IO board connections (if applicable)
- ☐ Inspect heat sinks and thermal pad/paste condition
- ☐ Check fan connectors and wiring
- ☐ Inspect daughterboards (if present)

4. Powered-On Diagnostics (Safe Mode)

- ☐ Connect power (not to mining pool)
- ☐ Observe startup sequence:
 - ☐ Fan startup and operation
 - ☐ LED indicator patterns
 - ☐ Listen for unusual sounds
 - ☐ Startup time normal/abnormal
- ☐ Record fan behavior:
 - ☐ All fans operating: Yes/No/Partial (list non-functioning fans)
 - ☐ Fans at proper speed: Yes/No
 - ☐ Unusual noise: Yes/No (describe: _____)
- ☐ Observe control board LED indicators:

- ☐ Power LED: On/Off/Flickering
- ☐ Status LED: On/Off/Flickering
- ☐ Network LED: On/Off/Flickering
- ☐ Hash board status LEDs: Notes: _____

5. Network Connectivity Check

- ☐ Connect the Ethernet cable
- ☐ Verify the link and activity LEDs are active
- ☐ Scan the network to find miner's IP or check the DHCP server
- ☐ Record IP address: _____
- ☐ Ping miner IP address: Success/Fail
- ☐ Check SSH access (for advanced diagnostics): Success/Fail
- ☐ Access web interface: Success/Fail
- ☐ Log in to web interface: Success/Fail
- ☐ Check for AUC detection (if applicable)

6. Web Interface/CGminer API Analysis

- ☐ Record temperature readings:
 - ☐ Control board: _____ °C
 - ☐ Hash board 1: _____ °C
 - ☐ Hash board 2: _____ °C
 - ☐ Hash board 3: _____ °C
- ☐ Check hash board detection:
 - ☐ Number of boards detected: _____/3
 - ☐ Chips detected per board: /
- ☐ Check system logs for errors:
 - ☐ Kernel log error messages: _____
 - ☐ Mining log error messages: _____
- ☐ Record system uptime: _____
- ☐ Verify current firmware version: _____
- ☐ Check power consumption data: _____ W
- ☐ Check chip frequency settings: _____ MHz
- ☐ Verify miner configuration parameters
- ☐ Check fan speed settings

7. Hash Board Diagnostics

Hash Board 1

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
 - ☐ Core voltage: _____ V (Target: Model-specific)
 - ☐ PLL voltage: _____ V (if applicable)
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _____ TH/s
- ☐ Operation stable for 10+ minutes: Yes/No
- ☐ Temperature within specs (<85°C typical): Yes/No
- ☐ Board-specific issues: _____
- ☐ Chain initialization successful: Yes/No/Partial

Hash Board 2

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
 - ☐ Core voltage: _____ V (Target: Model-specific)
 - ☐ PLL voltage: _____ V (if applicable)
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _____ TH/s
- ☐ Operation stable for 10+ minutes: Yes/No
- ☐ Temperature within specs (<85°C typical): Yes/No
- ☐ Board-specific issues: _____
- ☐ Chain initialization successful: Yes/No/Partial

Hash Board 3

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
 - ☐ Core voltage: _____ V (Target: Model-specific)
 - ☐ PLL voltage: _____ V (if applicable)
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _____ TH/s
- ☐ Operation stable for 10+ minutes: Yes/No
- ☐ Temperature within specs (<85°C typical): Yes/No
- ☐ Board-specific issues: _____
- ☐ Chain initialization successful: Yes/No/Partial

8. Avalon-Specific Diagnostic Tests

- ☐ Run built-in diagnostics through cgminer API if available
- ☐ Run manual PIC detection (if applicable)
- ☐ Check voltage regulators for proper operation
- ☐ Verify correct frequency configuration for model
- ☐ Test AUC connections and communication (older models)
- ☐ Check chain connection and continuity
- ☐ Test communication between the control board and hash boards
- ☐ Verify the chip initialization sequence is correct
- ☐ Test voltage stability under load

9. Performance Testing

- ☐ Connect to test mining pool (if authorized)
- ☐ Run for a minimum 30 minutes
- ☐ Record stable hash rate: _____ TH/s
- ☐ Record expected hash rate for model: _____ TH/s
- ☐ Performance percentage: _____% (Target: >95%)
- ☐ Record reject rate: _____% (Target: <1%)
- ☐ Record power consumption: _____ W
- ☐ Calculate efficiency: _____ J/TH
- ☐ Thermal stability: Pass/Fail
- ☐ Fan performance under load: Pass/Fail
- ☐ Network stability during test: Pass/Fail
- ☐ Monitor chip status during testing (through API)

10. Firmware Assessment

- ☐ Current firmware appropriate for model: Yes/No
- ☐ Current firmware version: _____
- ☐ Latest available firmware version: _____
- ☐ Firmware update required: Yes/No
- ☐ Recovery mode functioning: Yes/No
- ☐ Configuration settings appropriate: Yes/No
- ☐ Security settings configured properly: Yes/No

11. Model-Specific Checks

For Avalon 7 Series

- ☐ Check AUC module functionality
- ☐ Verify proper AUC device recognition
- ☐ Check per-chip frequency settings
- ☐ Verify voltage regulation per board

For Avalon 8/9 Series

- ☐ Check APW power supply compatibility
- ☐ Verify temperature sensor operation
- ☐ Test proper voltage offsets
- ☐ Check the fan control system

For Avalon 10/11 Series

- ☐ Verify cooling system operation
- ☐ Check the advanced power management system
- ☐ Test high-density board connections
- ☐ Verify firmware compatibility with hardware revision

12. Problem Diagnosis Summary

- ☐ Primary issue identified: _____
- ☐ Secondary issues identified: _____
- ☐ Root cause assessment: _____
- ☐ Components requiring repair/replacement:
 - ☐ _____
 - ☐ _____
 - ☐ _____

13. Repair Plan

- ☐ Recommended actions:
 - ☐ _____
 - ☐ _____
 - ☐ _____
- ☐ Estimated repair time: _____
- ☐ Parts required: _____
- ☐ Estimated cost: _____

☐ Repair priority: High / Medium / Low

14. Additional Notes

15. Final Recommendations

- ☐ Proceed with repairs
- ☐ Beyond economic repair - recommend replacement
- ☐ Partial repair recommended
- ☐ Additional diagnosis required
- ☐ Return to manufacturer (RMA)
- ☐ No issues found - return to service
- ☐ Firmware update only
- ☐ Preventative maintenance recommended

Diagnostic Completed By: _____

Signature: _____

Date/Time Completed: _____

Repair Approved By: _____

Date: _____

Avalon Error Code Reference

Error Code	Description	Possible Solution
Chain X: ASIC X: O	Open circuit	Check connections
Chain X: ASIC X: S	Short circuit	Inspect for shorts
Chain X: ASIC X: H	Overheating	Check cooling, thermal paste
Chain X: ASIC X: L	Low voltage	Check power supply, voltage reg
Chain X: PLL locktime failed	PLL issues	Check clock circuit, crystal
Voltage error	Incorrect voltage	Check PSU, voltage regulators
Failed to scan	Communication issue	Check data cables, control board
Chain X: Timeout	Chain not responding	Check connections, ASIC condition
Fan X: Error	Fan malfunction	Check fan connection, replacement
PIC connect timeout	PIC controller issue	Check PIC connection/programming